





National University of Science and Technology POLITEHNICA Bucharest
Faculty of Automatic Control and Computer Science
Department of Automatic Control and Systems Engineering

Preprocessing and Unsupervised Learning Methods for Electroencephalographic (EEG) Signal Analysis

Student: Maria-Daria Bălăucă

Supervisor: Associate Professor, PhD, Eng. Andra-Elena Băltoiu

ABSTRACT

The paper analyzes preprocessing and unsupervised learning methods for electroencephalographic (EEG) signals within two different experimental paradigms. The first study investigates cortical responses to affective visual stimuli, aiming to develop a real-time brain-computer interface. The results showed that extracting features from event-related potentials and removing artifacts lead to a better separation of EEG patterns compared to direct analysis of the continuous signal. The second study investigates brain dynamics in animal models in different pharmacologically induced states of consciousness. Based on a time-frequency representation of spectral power, cortical macrostates were identified, and their analysis highlighted a relationship between pre-stimulus brain activity and the subsequent response to stimulation. The obtained results allow the differentiation of levels of consciousness and highlight the potential of clustering methods in the computational analysis of neurophysiological states. Overall, the results can be used in the development of neuroadaptive systems and in the non-invasive monitoring of states of consciousness.